A TEN-VERTEX COUNTEREXAMPLE TO AN ENDPOINT DICHOTOMY FOR FACET IDEALS OF GRAPHIC MATROIDS, ANSWERING A QUESTION OF DIPASQUALE, FOULI, AND KUMAR

Abstract. DiPasquale, Fouli, and Kumar asked whether the asymptotic resurgence of the facet ideal of the graphic matroid of every simple 2-connected graph is always one of two extremal values. We give a ten-vertex graph W for which

$$c(W) = 8, \qquad \widehat{\alpha}(I(M(W))) = \frac{11}{2}, \qquad \widehat{\rho}(I(M(W))) = \frac{16}{9}.$$

Thus $2 - 2/c(W) = 7/4 < 16/9 < 9/5 = 2 - 2/|V(W)|$, answering their Question 7.2 in the negative.

For a connected graph G, let $M(G)$ be its graphic matroid and let $I(M(G)) \subseteq \mathbb{k}[x_e : e \in E(G)]$ be its facet ideal. Write $c(G)$ for the circumference of G. DiPasquale, Fouli, and Kumar proved that every simple 2-connected graph G satisfies

$$2 - \frac{2}{c(G)} \leq \widehat{\rho}(I(M(G))) \leq 2 - \frac{2}{|V(G)|},$$

the upper bound being [1, Theorem 5.5], and asked whether the asymptotic resurgence is always one of the two endpoints [1, Question 7.2]. Only the upper bound is used below; the lower bound is quoted here as context for the question.

Let W be the graph on $\{0, 1, \dots, 9\}$ with

$$E(W) = \{05, 15, 06, 26, 17, 27, 08, 38, 48, 58, \\ 29, 39, 49, 79\}. \tag{1}$$

Here ij abbreviates the edge $\{i, j\}$. Its graph6 code is `I?BDBA[MO`. Deleting vertex 3 gives the graph in [1, Example 6.1] under $(v_1, w_1, u_1, p, q, r, v_2, w_2, u_2) = (8, 5, 0, 4, 6, 1, 9, 7, 2)$; thus W is obtained from that graph by duplicating one of its degree-two vertices, p.

Theorem 1. *The graph W is simple and 2-connected, and*

$$c(W) = 8, \qquad \widehat{\alpha}(I(M(W))) = \frac{11}{2}, \qquad \widehat{\rho}(I(M(W))) = \frac{16}{9}.$$

Consequently, W is a counterexample to the endpoint dichotomy in [1, Question 7.2].

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Proof. The graph is simple, and

$$0-5-1-7-2-6-0, \quad 0-8-5, \quad 2-9-7, \quad 8-3-9, \quad 8-4-9$$

is an open-ear decomposition, so W is 2-connected.

Vertices 3 and 4 are nonadjacent twins with neighborhood $\{8,9\}$. Any cycle containing both is therefore the 4-cycle $8-3-9-4-8$, so W has no 10-cycle.

A 9-cycle would omit exactly one of $3,4$; by symmetry, suppose it omits 4. In $W-4$, the degree-two vertices $1,3,6$ force the paths

$$5-1-7, \qquad 8-3-9, \qquad 0-6-2.$$

The remaining cycle edges would form a perfect matching on their six endpoints. The available edges on those endpoints form the disjoint triangles

$$W[\{0,5,8\}] \cong K_3, \qquad W[\{2,7,9\}] \cong K_3,$$

which have no perfect matching. Hence no 9-cycle exists. Since

$$5-0-8-3-9-2-7-1-5$$

is an 8-cycle, $c(W)=8$.

We next compute the Waldschmidt constant. For $x \in \mathbb{R}_{\geq 0}^{E(W)}$ and $F \subseteq E(W)$, put $x(F) = \sum_{e \in F} x_e$. The bond description of the symbolic polyhedron and its linear-programming formula give

$$\widehat{\alpha}(I(M(W))) = \min\{x(E(W)) : x(B) \geq 1 \text{ for every bond } B \text{ of } W\} \tag{2}$$

[1, Corollary 2.2 and Theorem 2.3].

Because W is 2-connected, every incident edge cut $\delta(v)$ is a bond. Put

$$S = \{3,4,8,9\}, \qquad B_0 = \delta(S) = \{08, 58, 29, 79\}.$$

Both $W[S]$ and $W[V(W) \setminus S]$ are cycles, so B_0 is a bond. Every edge of W occurs exactly twice in the following weighted sum of bond inequalities. Thus every feasible x in (2) satisfies

$$\begin{aligned} 2x(E(W)) &= \sum_{v \in \{0,1,2,5,6,7\}} x(\delta(v)) + 2x(\delta(3)) + 2x(\delta(4)) + x(B_0) \\ &\geq 11. \end{aligned}$$

Hence $\widehat{\alpha}(I(M(W))) \geq 11/2$.

For the reverse inequality, let J be the spanning subgraph formed by the two cycles

$$8-0-6-2-7-1-5-8 \qquad \text{and} \qquad 8-3-9-4-8,$$

of lengths 7 and 4. The graph J is connected and bridgeless and has $7+4=11$ edges. Set $x_e = 1/2$ for $e \in E(J)$ and $x_e = 0$ otherwise. If $B = \delta_W(A)$ is a bond of W, then

$$B \cap E(J) = \delta_J(A).$$

Since J is connected and bridgeless, $|\delta_J(A)| \geq 2$; hence $x(B) \geq 1$. This x is feasible in (2) and has total weight $11/2$. Therefore

$$\widehat{\alpha}(I(M(W))) = \frac{11}{2}. \tag{3}$$

By the identification above,

$$M(W-3) = M(W) \setminus \{38, 39\}.$$

The graph $W-3$ is the graph of [1, Example 6.1], whose facet ideal has asymptotic resurgence $16/9$. Minor monotonicity [1, Proposition 2.4] therefore gives

$$\widehat{\rho}(I(M(W))) \geq \frac{16}{9}. \tag{4}$$

Since $|V(W-3)| = 9$, the value $16/9$ equals $2 - 2/|V(W-3)|$: the graph of [1, Example 6.1] sits at the upper endpoint of its own bound and is not itself a counterexample to Question 7.2.

Since $M(W)$ is simple of rank 9, the contraction formula [1, Theorem 2.5] yields

$$\widehat{\rho}(I(M(W))) = \max\left\{\frac{9}{\widehat{\alpha}(I(M(W)))}, \max_{e \in E(W)} \widehat{\rho}(I(M(W)/e))\right\}. \tag{5}$$

The first term is $18/11 < 16/9$ by (3).

We use the following consequence of the reduction results in [1]: if N is a loopless graphic matroid of rank r, then

$$\widehat{\rho}(I(N)) \leq 2 - \frac{2}{r+1}. \tag{6}$$

Indeed, simplification preserves asymptotic resurgence, and the asymptotic resurgence of a direct sum is the maximum over its matroid-connected components [1, Theorems 3.2 and 3.3]. A rank-one component contributes 1. A simple connected graphic component of rank $s \geq 2$ is represented by a simple 2-connected graph on $s+1$ vertices, so [1, Theorem 5.5] gives the bound $2 - 2/(s+1) \leq 2 - 2/(r+1)$.

For each $e \in E(W)$, the matroid $M(W)/e$ is loopless, graphic, and of rank 8. Hence (6) gives

$$\widehat{\rho}(I(M(W)/e)) \leq \frac{16}{9}.$$

Equation (5) now gives the reverse inequality to (4), and therefore

$$\widehat{\rho}(I(M(W))) = \frac{16}{9}.$$

The claimed strict inequalities follow from $c(W) = 8$ and $|V(W)| = 10$. $\square$

Remark. DiPasquale, Fouli, and Kumar record that they expect a negative answer to Question 7.2, while their Sage computations indicate that the answer is yes for every 2-connected graph on nine vertices or less [1, Table 1 and the discussion following Question 7.2]. The content of Theorem 1 is therefore the explicit ten-vertex witness, not a reversal of the source authors' expectation. Concretely, duplicating p raises the order from 9 to 10 and leaves the asymptotic resurgence at 16/9, while the upper endpoint moves from $2-2/9 = 16/9$ to $2-2/10 = 9/5$; the value that is extremal for $W-3$ is therefore strictly interior for W. Conditional on their computation, W has minimum possible order; we do not repeat that enumeration here. Nothing above determines which values other than 16/9 occur strictly between the two endpoints, nor whether W is the only ten-vertex example.

Remark. Every numbered result quoted above is numbered as in [1], and several of them originate elsewhere. The linear-programming formula for the Waldschmidt constant, [1, Theorem 2.3], is stated there with attribution to Cooper, Embree, Hà, and Hoefel. The contraction formula, [1, Theorem 2.5], is stated there with the citation [2, Proposition 3.8 and Theorem 3.9] attached, [2] being the same three authors' companion paper on matroids; minor monotonicity, [1, Proposition 2.4], is by contrast a result of [1] itself, proved there from [2, Corollary 3.4 and Remark 3.7]. The remaining items used above, namely [1, Corollary 2.2], [1, Theorems 3.2, 3.3 and 5.5], [1, Example 6.1] and [1, Table 1], are from [1] itself.

References

[1] M. DiPasquale, L. Fouli, and A. Kumar, *Asymptotic resurgence of facet ideals of graphic matroids*, arXiv:2607.26294v1 [math.AC], 2026.

[2] M. DiPasquale, L. Fouli, and A. Kumar, *Asymptotic resurgence of facet and Stanley–Reisner ideals of matroids*, arXiv:2607.20892 [math.AC], 2026.